1.6 Electrochemistry

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.6 Electrochemistry
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

When heated ammonium nitrate, NH_4NO_3, can decompose explosively.

$$NH_4NO_3 \rightarrow N_2O + 2H_2O$$

The nitrogen atoms in NH_4NO_3 have different oxidation numbers.

What are the changes in the oxidation numbers when this reaction proceeds?

A. +4, -4

B. -2, -4

C. +4, -6

D. +2, +6

[1 mark]

Question 2

The acid HIO disproportionate when in aqueous solution, this is shown in the equation, where q, r, s and t are simple whole numbers in their lowest ratios.

$$\mathbf{q}\,HIO \rightarrow \mathbf{r}\,I_2 + \mathbf{s}\,HIO_3 + \mathbf{t}\,H_2O$$

The equation can be balanced using oxidation numbers.

What are the values for r and s?

	r	**s**
A	4	2
B	4	1
C	2	1
D	1	2

[1 mark]

Question 3

In winemaking to prevent oxidation of ethanol by air, sulfur dioxide (SO_2) is added. In order to calculate the amount of SO_2, a sample is titrated with iodine (I_2). The reaction is a one to one ratio for SO_2 and I_2.

What is the change in the oxidation number of sulfur in this reaction?

A. +2 to +6

B. +4 to +6

C. +2 to +4

D. +4 to +5

[1 mark]

Question 4

The table shows the redox potentials of a number of species.

Half reaction	$E^{\ominus}$ / V
$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	- 0.26
$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+ 0.34
$S_4O_6^{2-}(aq) + 2e^- \rightleftharpoons 2S_2O_3^{2-}(aq)$	+ 0.47
$2VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	+ 1.00

What is the lowest possible oxidation number that Vanadium (V) can be reduced to by thiosulphate ions?

A. +2

B. +3

C. +4

D. +5

[1 mark]

Question 5

The oxidation state of chlorine can range from –1 to +7 in its compounds.

Which reagent(s) and conditions are necessary to oxidise elemental chlorine into a compound containing chlorine in the +5 oxidation state?

A. Concentrated H_2SO_4 at room temperature.

B. $AgNO_3$ (aq) followed by NH_3 (aq) at room temperature.

C. Cold dilute NaOH (aq).

D. Hot concentrated NaOH (aq).

[1 mark]

Question 6

A student set up experiments to observe the outcome of adding different sodium halides to concentrated phosphoric acid and concentrated sulfuric acid. She recorded her findings in the table below.

	sodium chloride	**sodium iodide**
Conc. H_3PO_4	Colourless acidic gas formed	Colourless acidic gas formed
Conc. H_2SO_4	Colourless acidic gas formed	Purple vapour formed

Which deduction can be made from these observations?

A. Concentrated sulfuric acid is a stronger oxidising agent than chlorine.

B. Concentrated sulfuric acid is a stronger oxidising agent than iodine.

C. Concentrated phosphoric acid is a stronger oxidising agent than iodine.

D. Concentrated phosphoric acid is a stronger oxidising agent than concentrated sulfuric acid.

[1 mark]

Question 7

If a solution contains both bromine and chlorine, BrO_3^- ions are produced.

The reactions leading to the production of BrO_3^- ions are shown below:

Reaction 1: $Br_2 + H_2O \rightarrow HBr + HBrO$

Reaction 2: $3HBrO + Cl_2 \rightarrow 2Cl^- + BrO_3^- + Br_2 + 3H^+$

1 Chlorine is reduced in reaction 2.

2 Bromine is reduced in both reaction 1 and reaction 2.

3 Bromine is oxidised in both reaction 1 and reaction 2.

Which statements about these reactions are correct?

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 8

If a dilute acid is added to an aqueous solution containing nitrite ions, NO_2^-, two different nitrogen compounds are released as gases.

$$2H^+(aq) + 2NO_2^-(aq) \rightarrow H_2O(l) + NO(g) + NO_2(g)$$

Which of the three statements below correctly describe the process?

1 The H^+(aq) ion is oxidised by NO_2^-(aq).

2 Some nitrogen atoms are oxidised, and some nitrogen atoms are reduced.

3 The H^+(aq) ion acts as a catalyst.

A. 1 and 2

B. 2 only

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 9

Which pair of reactants will result in a redox reaction?

1 CH_3COCH_3 + Tollens' reagent

2 CH_3CH_2CHO + Fehling's reagent

3 $C_2H_4 + Br_2$

A. 1 only

B. 1 and 2

C. 2 and 3

D. 3 only

[1 mark]

Question 10

How does concentrated sulfuric acid behave when it reacts with sodium chloride?

A. As a reducing agent only.

B. As an oxidising agent only.

C. As an acid only.

D. As an acid and an oxidizing agent.

[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- In the NH_4^+ ion, the oxidation state of nitrogen is -3.
- In the NO_3^- ion the oxidation state of nitrogen is +5.
    - Both of the nitrogen atoms having a different oxidation state.
- In N_2O the oxidation state of oxygen is always -2, so each nitrogen must have an oxidation state of +1.
- The change in oxidation state is, therefore:
    - From -3 in the NH_4^+ ion to +1 in the N_2O compound a change of +4.
    - From +5 in the NO_3^- ion to +1 in the N_2O compound a change of -4.

2

The correct answer is **C** because:

- In this reaction, the only atom to have a change of oxidation state is the iodine.
- In HIO iodine has an oxidation state of +1 in I_2 it is 0.
- In HIO_3 the oxidation state of iodine is +5.
- For the reactants and products to be balanced the coefficient of HIO is 5.
- Balancing the equation:
    - $5HIO \rightarrow 2I_2 + HIO_3 + 2H_2O$

A, **B** & **D** are incorrect as these options do not correctly balance the equation.

3

The correct answer is **B** because:

- The equation for the reaction will be:
  - $2H_2O + SO_2 + I_2 \rightarrow 2HI + H_2SO_4$
- The oxidation state of sulfur in SO_2 is **+4** as each oxygen has an oxidation state of -2.
- The oxidation state of sulfur in H_2SO_4 is **+6**; each hydrogen is +1, and there are 4 oxygen atoms each with -2.

A, **C** & **D** are incorrect as these answers do not correspond to the actual changes in oxidation number involved.

4

The correct answer is **C** because:

- Vanadium (V) is present as the ion VO_2^+.
- For a feasible reaction to take place $E_{cell}^{\varnothing}$ (or the emf) must be positive.
- To act as a reducing agent thiosulphate ions, $S_2O_3^{2-}$, must provide electrons and the half equation will go backwards.
  - $2S_2O_3^{2-} \rightleftharpoons S_4O_6^{2-}(aq) + 2e^-$
- For this to be the case the thiosulphate half reaction potential must be more negative than the reaction containing the vanadium species.
- Only a reaction from Vanadium (V) to Vanadium (IV) will give a positive emf.
  - $4VO_2^+ + 4H^+ + 2S_2O_3^{2-} \rightleftharpoons 2VO^{2+} + 2H_2O + S_4O_6^{2-}$ $E_{cell}^{\varnothing} = 1.00 - 0.47 = +0.53$ V
  - $2VO^{2+} + 4H^+ + 2S_2O_3^{2-} \rightleftharpoons 2V^{3+} + 2H_2O + S_4O_6^{2-}$ $E_{cell}^{\varnothing} = 0.34 - 0.47 = -0.13$ V
  - $2V^{3+} + e^- + 2S_2O_3^{2-} \rightleftharpoons 2V^{2+} + S_4O_6^{2-}$ $E_{cell}^{\varnothing} = -0.26 - 0.47 = -0.73$ V
- The final oxidation number is therefore +4.

A, B & **D** are incorrect as these answers do not correspond to the actual changes in oxidation number involved.

5

The correct answer is **D** because:

- The equation for the reaction is:
  - $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClCO_3 + 3H_2O$
  - The oxidation state of chlorine in $NaClCO_3$ is +5.

- Sodium has an oxidation state of +1.

- Oxygen has a total of -6.
  - $(+1) + (x) + (-6) = 0$
  - Therefore, $x = 6 - 1 = +5$

A is incorrect as there is no reaction.

B is incorrect as in the first reaction, it will form AgCl in which chlorine has a -1 charge, it is then followed by aqueous ammonia and will produce a ligand in which chlorine still has a -1 charge, $[Ag(NH_3)_2]^+ Cl^-$.

C is incorrect as when elemental chlorine reacts with cold NaOH the overall reaction is: $2NaOH + Cl_2 \rightarrow NaCl + NaClO + H_2O$. There is no product here that has chlorine in the +5 oxidation state. In NaCl the chlorine has an oxidation state of -1, and in the NaClO it has an oxidation state of +1.

6

The correct answer is **B** because:

- Sulfuric acid can act as both an acid and oxidising agent when mixed with halide ions.

- The product of this reaction will be hydrogen halide fumes.

- The iodide ions are oxidised to iodine (the purple vapour formed) by the concentrated sulfuric acid.
  - $2I^- \rightarrow I_2 + 2e^-$

- The iodide ions are strong enough reducing agents to reduce the sulfuric acid.
  - Firstly to sulfur dioxide (sulfur oxidation state +4)
  - Then to sulfur (oxidation state 0)

- Then to hydrogen sulphide (sulfur oxidation state -2)

A is incorrect as concentrated sulfuric acid is not a strong enough oxidising agent to

  - $2V^{3+} + e^- + 2S_2O_3^{2-} \rightleftharpoons 2V^{2+} + S_4O_6^{2-}$ $E_{cell}^{\ominus} = -0.26 - 0.47 =$ -0.73 V

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- Sodium has an oxidation state of +1.
- Oxygen has a total of -6.
  - $(+1) + (x) + (-6) = 0$
  - Therefore, $x = 6 – 1 = +5$

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    - Then to sulfur (oxidation state 0)
    - Then to hydrogen sulphide (sulfur oxidation state -2)

A is incorrect as concentrated sulfuric acid is not a strong enough oxidising agent to oxidise chloride ions.

C & **D** are incorrect as phosphoric acid is not an oxidising agent.

7

The correct answer is **D** because:

- **Statement 1** states that chlorine is reduced in reaction 2, this is correct as the oxidation state of chlorine has gone from 0 to -1, gaining electrons and is reduced.

- **Statement 2** states that bromine is reduced in reaction 1 and 2.
    - **Reaction 1**, bromine goes from an oxidation state of 0 in Br_2 to -1 in HBr so is reduced (gained electrons).
    - **Reaction 2**, bromine goes from an oxidation state of +1 in HBrO to 0 in Br_2 so is reduced (gained electrons).

- **Statement 3** states that bromine is oxidised in reaction 1 and 2.
    - **Reaction 1,** bromine goes from an oxidation state of 0 in Br_2 to +1 in HBrO so is oxidised (lost electrons).
    - **Reaction 2**, bromine goes from an oxidation state of +1 in HBrO to +5 in BrO_3^- so is oxidised (lost electrons)

8

The correct answer is **B** because:

- Nitrogen in NO_2^- has the oxidation state of +3, each oxygen atom has an oxidation state of -2 overall charge of the ion is -1 so nitrogen must be +3.

- In NO it has an oxidation state of +2, so has been reduced from +3 to +2 (gained electrons).
- In NO_2 nitrogen has an oxidation state of +4 so has been oxidised from +3 to +4 (lost electrons).

A is incorrect as the hydrogen ions are not oxidised as the oxidation state stays the same.

C & **D** are incorrect as a catalyst is not used in the reaction. The H^+ ions are used in the reaction, so it is not acting as a catalyst.

9

The correct answer is **C** because:

- CH_3CH_2CHO is an aldehyde (propanal) and will be oxidised by Fehling's reagent to produce ethanoic acid.
- Fehling's solution is reduced by propanal; this is, therefore, a redox reaction.
- The oxidation state of each carbon atom in C_2H_4 is -2, when aqueous bromine is added electrophilic addition occurs across the double bond, and the oxidation number increases to -1.
- The bromine atoms are reduced from 0 to -1; this is, therefore, a redox reaction.

There is no reaction with Tollens' reagent and a ketone (in this case acetone). Tollens' reagent has a redox reaction with aldehydes (silver mirror test).

10

The correct answer is **C** because:

- Sulfuric acid can act as both an acid and oxidising agent when mixed with halide ions.
- Concentrated sulfuric acid is not a strong enough oxidising agent to oxidise chloride ions.
- Therefore with chloride ions, it will react as an acid only.